CIPA target prioritisation — project walkthrough

Oct 10, 2026 · @Simone Edoardo

The T-dark exam project on CIPA has been rebuilt as a reproducible pipeline that proposes and ranks candidate therapeutic targets for a rare monogenic disease. The headline result is not the ranking: it is that the best-supported targets serve the fewest patients, and that the two genuinely understudied candidates are understudied partly because the tissue they act in has never been measured at scale.

From the exam to the pipeline

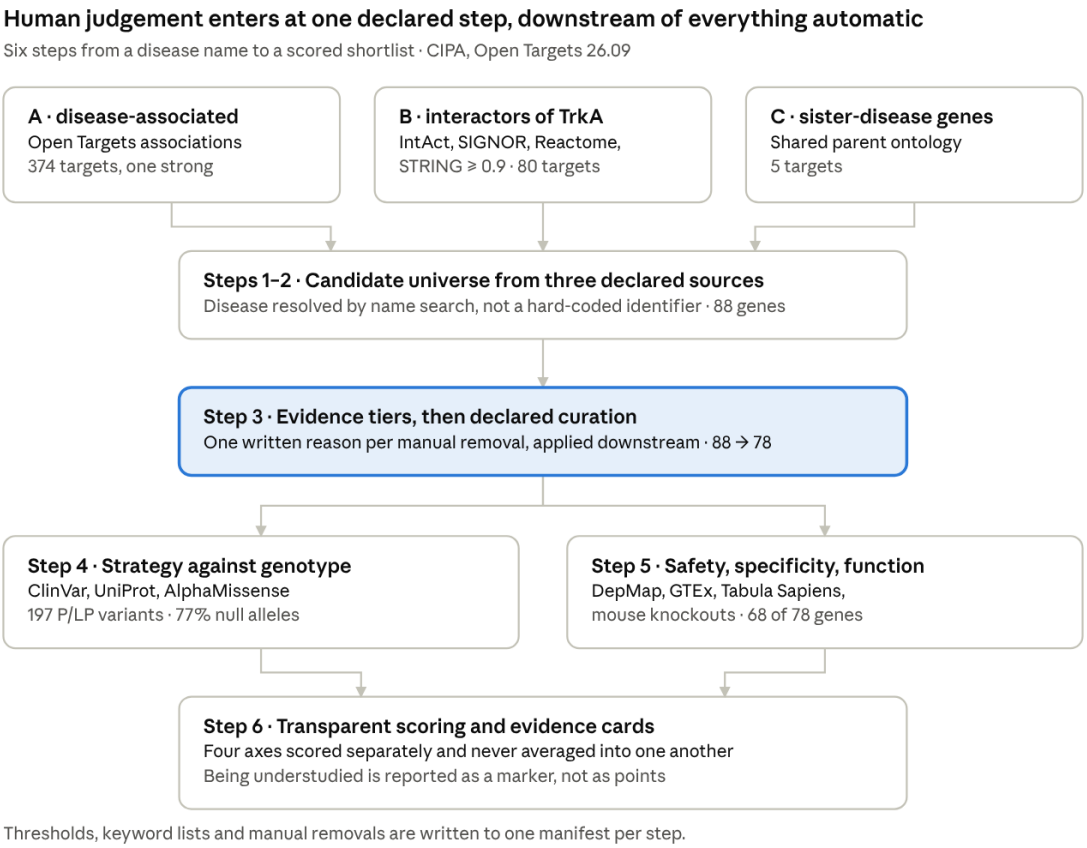
The exam project asked which T-dark genes associated with CIPA might serve as therapeutic targets, with reference to the in vivo and in vitro models already in the literature. Selection was manual: 36 genes returned by Open Targets, filtered by no known drugs, no chemical probe and at most three tractability assessments, leaving nine, of which NTF3 and PRDM12 were developed in depth.

Re-running that first step on the current Open Targets release (26.09) returns **374** associated genes rather than 36. Only *NTRK1* carries a strong association (0.86) backed by genetic evidence. The rest is a long tail of mouse-phenotype similarity, literature co-mention and the targets of a symptomatic drug.

That changed the framing. In a monogenic loss-of-function disease, an association list contains one cause and a weak tail, so filtering that tail for T-dark genes selects genes that are obscure but only loosely tied to the disease. The plausible targets sit in the biology **around** the causal gene, which is why two further candidate sources were added.

The T-dark criterion was kept, but split in two. The exam criterion selects genes with **no chemical matter**; the Pharos development level selects genes that are **poorly studied**. These are different axes, and all eight exam genes still associated turn out to be Tbio, none Tdark — so the criterion was not wrong, it was measuring the other axis.

How the pipeline works



pipeline overview · 3 candidate sources, 6 steps

Three declared sources feed one candidate universe. Everything above Step 3 is automatic; every human judgement enters inside Step 3 and nowhere else. Steps 4 and 5 ask independent questions — which strategy a genotype permits, and whether a gene is safe and in the right tissue — and only meet in the scoring step.

Steps 1–2 — From a disease name to a candidate universe

The disease is resolved by **name search**, not by a hard-coded identifier, so the same code transfers to another disease. CIPA resolves to MONDO_0009746, hereditary sensory and autonomic neuropathy type 4. Every associated target is retrieved with its evidence broken down by type, and the Open Targets release is written into a manifest.

Two further sources are then added, and each gene keeps a label saying where it came from:

Source	Definition	Genes
A disease-associated	Step 1, weighted by evidence type	374

Source	Definition	Genes
B mechanistic neighbours	direct interactors of the causal gene: IntAct ≥ 0.6, SIGNOR, Reactome, STRING ≥ 0.9	80
C sister-disease genes	causal genes of diseases sharing a parent class in the ontology	5

Source C needed no manual choice. The ontology parent *congenital insensitivity to pain* has seven children, and its other members returned *PRDM12*, *SCN9A*, *SCN11A*, *NGF* and *ZFHX2* — two developmental transcription factors and two sodium channels, which is a clean split between genes that build nociceptors and genes that make them fire.

The most informative number here is that sources A and B barely overlap: only six genes appear in both. The disease-association list does not contain the TrkA signalling machinery, which is exactly what source B was added to recover.

Step 3 — Evidence tiers, and curation kept separate from computation

Candidates are tiered by how good the evidence behind them is: seed, sister-disease gene, curated interaction, STRING-only interaction, disease association with genetic or clinical support, and background. Only the first five form the shortlist pool.

One automatic check earns its place. Genetic evidence on a gene lying within 1 Mb of the causal gene may belong to the causal locus rather than to that gene. The check flagged *INSRR* and *SH2D2A*, both immediately adjacent to *NTRK1* at 1q23 — and *SH2D2A* was one of the nine genes selected in the exam.

The part I would most like your opinion on is structural. **Every manual judgement is made in one declared block, each with a written reason, applied on top of the automatic output rather than inside it.** Step 3 writes `02b_universe_tiered.csv` from computation alone; the curation step reads that file and writes `02c_*`. Anyone can diff the two and see precisely what a human decided.

What is declared there: partners of oncogenic *NTRK1* fusions (*TPM3*, *TPR*, *TFG*) and ligands of unrelated receptor tyrosine kinases linked only by STRING, both flagged out of the pool; a minimum genetic score for genes supported by source A alone; and mechanism classes grounded in TrkA biology rather than in generic protein-family labels. The pool goes from 88 to **78**.

Step 4 — Which strategy can serve which genotype

A strategy only helps patients whose variant leaves room for it, so the pathogenic variants of the causal gene were characterised before any candidate was ranked. ClinVar returns 1,859 records for *NTRK1*; restricted to single-gene variants on the reference transcript NM_002529.4 this gives 1,530, of which **197 are pathogenic or likely pathogenic**.

Before trusting any of it, the reference residue of every missense variant was checked against the UniProt sequence for P04629. All of them match, so positions, domains and AlphaMissense scores are on the same coordinate system.

The distribution decides the project's main conclusion. **77% are null** — 76 frameshift, 41 nonsense, 35 canonical splice — and a further 26 are missense inside the kinase domain. Roughly nine variants in ten leave no functional TrkA kinase. Only six missense variants sit in the extracellular region, with two in-frame indels.

It follows that any strategy acting **through** TrkA — supplying ligand, raising expression, transactivating the receptor — can serve at most a few per cent of variants. Strategies that bypass TrkA, and restoring the gene itself, are the only broadly applicable routes.

AlphaMissense was used on the missense set and agrees with ClinVar on 30 of 32 pathogenic and 22 of 22 benign variants, which is the check that it behaves sensibly on this protein. It also calls 83 of 397 variants of uncertain significance likely pathogenic — a prioritised list for reclassification, not a reclassification. One further correction to the exam write-up: per-residue intolerance is far higher in the kinase domain (mean 0.70) than in the extracellular region (0.38), whereas the exam described the NGF-binding region as the most susceptible.

A companion step reads each variant's HGVS string and asks what kind of edit would revert it: 31% transitions, 24% transversions, 41% small indels, 3% large. Since no variant dominates, a per-variant correction approach needs one guide per variant, while gene addition is a single construct for every genotype. This is notation-level only — no PAM sites, editing windows, guides, off-target or efficiency assessment.

Step 5 — Safety, tissue specificity, and the one orthogonal axis available

Up to this point every piece of evidence came, directly or indirectly, from the same body of literature and curation that defines the disease. Step 5 adds the things that can disqualify a target, and the one data type that was generated for a different purpose entirely.

Safety and essentiality

For each of the 78 candidates the pipeline retrieves DepMap gene-effect scores through Open Targets, together with curated cancer roles and known safety liabilities. Two declared thresholds: a mean gene effect below **−0.5** marks a dependency, below **−1.0** a strong one. A gene that most cancer cell lines cannot survive without is a poor candidate for systemic activation or inhibition, whichever direction the strategy needs.

The cancer-role flag was initially returning zero for the whole pool, which was obviously wrong — *PIK3CA*, *BRAF* and *SRC* are all in it. The cause was an assumption about the vocabulary used in the `hallmarks.attributes` field. Printing the distinct attribute names (`role in cancer`, `fusion partner`, `types of alteration in cancer`, …) and matching on them fixed it. This is recorded here because it is the kind of silent zero that a pipeline will happily report forever.

Tissue specificity read against two resources that ask different questions

The obvious thing to do with expression data is to ask whether a candidate is enriched in nervous tissue. The pipeline asks it twice, of two resources, and treats the *disagreement* as the informative part.

- **Bulk GTEx** asks whether a gene concentrates in nervous **tissue**.
- **Tabula Sapiens single-cell** asks whether it concentrates in neural **cell types**.

A sensory neuropathy is a disease of a sparse, scattered cell population. A gene expressed only in nociceptors is diluted to nothing in a bulk tissue homogenate. So the four possible outcomes are not four grades of the same thing; they are four different situations:

pattern	reading
both	the signal survives either framing — the strongest case
single-cell only	sparse cell type diluted away in bulk; for this disease the *expected* pattern, and where *NTRK1*, *NGF* and *SCN9A* land
bulk only	present in nervous tissue, but perhaps not in neurons — a question, not a score
not assessable	below detection in both atlases

Two measurement details matter, and both were mistakes before they were decisions.

First, **the tissue list is declared, not matched by keyword**. An early version matched tissue names containing "cortex", "nerve", "gangli" and similar. In the GTEx vocabulary this recognised `cortex of kidney` as nervous tissue and missed most of the brain. Every enrichment figure in that run was meaningless. The pipeline now carries an explicit list of GTEx biosample names, read off a printed inventory of what the resource actually returns, and raises an error if a declared name is absent from the response — so a vocabulary change in a future Open Targets release fails loudly instead of quietly.

Second, **enrichment is a fold-change against the gene's own median across all tissues, with a pseudocount**:

```
PSEUDOCOUNT = 1.0   # TPM
nervous_fold = (max_nervous + PSEUDOCOUNT) / (median_all_tissues + PSEUDOCOUNT)
```

Without the pseudocount, the most brain-specific genes in the pool — the ones the measure exists to find — returned `NaN`, because a gene silent in 45 of 50 tissues has a median of exactly zero. *RIT2* was the case that exposed it. Note also what the measure is *not*: it is not the nervous share of a gene's own maximum, which would score a gene expressed nowhere at all as perfectly nervous-specific.

Mouse knockout phenotypes

This is the only axis in the whole pipeline that is functional rather than associative, and it comes from experiments nobody performed with CIPA in mind. Phenotypes are retrieved for each candidate and sorted into three declared tiers:

- **A** — nociception, sensory ganglia, sweat glands
- **B** — peripheral nerve, axons, autonomic function, thermoregulation
- **C** — skin and appendages

68 of the 78 candidates have a mouse model. Here too the first implementation was too permissive: keywords like `gangli`, `sensory`, `axon` and `innervation` matched retinal, cochlear, vestibular, motor and enteric phenotypes, and *SORT1* scored a full tier-A match on a **retinal** phenotype. A

`NOT_SOMATOSENSORY` exclusion list now removes 15 such records before tiering. The exclusions are declared in the manifest, which means a reader who thinks `enteric` belongs in tier B can say so and change one line.

The result is reported in the next step, but it is worth naming the limitation now: for the five genes that entered the universe as *sister-disease causal genes* (source C), a pain phenotype in the mouse is not independent of the reason they were selected. The axis is genuinely new information for source B — *KIDINS220*, *RAB7A*, *NTF4*, *NGFR*, *GDNF*, *NTRK2*, *ADORA2A* — which entered on interaction evidence alone.

Step 6 — Scoring that can be disagreed with, and an evidence card per candidate

The scoring step is deliberately the least clever part of the pipeline. Four axes, declared weights, integer points, **no normalisation and no averaging of one axis into another**:

axis	max	what it measures
disease link	3	strength and type of evidence connecting the gene to the disease
tissue specificity	2	the two-resource pattern from Step 5
safety	2	absence of essentiality, cancer role, known liabilities
mouse knockout	3	tier of the most relevant knockout phenotype

Total 10. Each candidate's row carries the four components alongside the total, so a reader who thinks the mouse axis is worth 1 point rather than 3 can recompute the ranking in their head.

Being understudied is reported as a marker, not as points. This is the single most important scoring decision in the project, and it is the one that departs most from the original exam brief. A Pharos Tdark label tells you that nobody has worked on a gene. That is an *opportunity* — a reason a PhD project might be worth doing — but it is not evidence that the gene matters in the disease. Adding it to the score would mean rewarding ignorance. It therefore sits in its own column, next to a separate column for how much chemical matter exists, because "no known drug" and "poorly understood" are different axes and the exam criterion conflated them.

Strategy and direction

A score without a mechanism is not actionable, so every candidate also carries three declared fields: the **therapeutic strategy** it implies, the **direction** the target would have to be moved, and which **variant groups** from Step 4 that strategy could serve.

Direction matters more than it sounds. CIPA is a loss-of-function disease, so almost every strategy requires *activation* — and most existing pharmacology does the opposite. The clearest case: *SCN9A* has

79 clinical-stage molecules behind it, every one developed to **block** the channel for analgesia. In this disease that is precisely backwards, and the pipeline flags it rather than counting the 79 molecules as tractability.

Strategy assignment is mostly by mechanistic class, with a declared per-gene override where the class would be biologically wrong. The override exists because the first version mapped every "neurotrophin ligand or co-receptor" to "raise signalling through TrkA" — which is false for *GDNF* (RET/GFRα, a parallel pathway), for *BDNF* and *NTF4* (parallel Trk receptors), and unclear for *NGFR* and *SORT1*, where the direction itself has to be established first.

The shortlist

gene	source	strategy	direction	score	note
SCN9A	C	restore excitability of surviving neurons	activate	10	excellent target, weakest strategy
ZFHX2	C	redirect nociceptor specification	activate	10	**Tdark**, no chemical matter
SCN11A	C	restore excitability of surviving neurons	activate	10	as *SCN9A*
NGF	C	raise signalling through TrkA	activate	10	serves ~4% of variants
NTRK1	seed	restore *NTRK1* itself	restore / replace	8	the only broadly applicable route
PRDM12	C	redirect nociceptor specification	activate	8	understudied; specificity not assessable
KIDINS220	B	re-engage the pathway below the receptor	activate	8	understudied
GDNF	B	recruit a parallel neurotrophic pathway	activate	8	does not use TrkA
NTF3	B	raise signalling through TrkA	activate	8	understudied

gene	source	strategy	direction	score	note
NGFR	B	modulate the p75 co-receptor	**unclear**	8	direction must be established first

Evidence cards

The final output per candidate is a one-page card: every retrieved field, the source it came from, the date of retrieval, and the declared thresholds that produced each flag. The point of the card is that no number in it is a black box — a reader can trace any value back to the API response cached in `data/raw/`. For a project whose conclusions cannot be validated experimentally, traceability is the only form of accountability available.

What the pipeline found

Four results, in descending order of how much I trust them.

1. It recovers a ground truth it was never given

Six genes are known to cause congenital insensitivity to pain in humans: *NTRK1*, *NGF*, *SCN9A*, *SCN11A*, *PRDM12*, *ZFHX2*. Those six are **exactly** the six candidates in the pool whose mouse knockouts produce a tier-A nociception phenotype, and they occupy the top of the ranking.

Mouse phenotype data enters the pipeline through a separate query, from a resource built for a different purpose, and plays no part in how candidates were selected. So this is a positive control rather than a restatement — with the caveat that for the four source-C genes, their pain phenotype is not fully independent of the reason they were included. The control is cleanest for *NTRK1* and *NGF*.

2. The best-supported targets serve the fewest patients

This is the result I would lead with in a discussion, because it is the one that changed my view of the problem.

Of 197 pathogenic / likely pathogenic *NTRK1* variants in ClinVar, **77% are null** (nonsense, frameshift, splice, large deletion) and a further 13% are kinase-domain missense. Roughly **nine in ten patients have no functional TrkA kinase**. Every strategy that acts *through* the receptor — supplying NGF, raising *NTRK1* expression, transactivating TrkA from a GPCR — can therefore serve only a few per cent of the variant spectrum. Yet those strategies score highest, because they are the ones with ligands, mouse phenotypes and chemical matter behind them.

The broadly applicable route — restoring *NTRK1* itself — is a single row with almost nothing to score.

The pipeline reports this tension rather than resolving it. A ranking that smoothed it away by folding genotype coverage into the total would have produced a tidier table and a worse result.

3. The two genuinely dark candidates are dark because the tissue was never measured

Of 78 curated candidates, only **PRDM12** and **ZFHX2** fall in the understudied-and-unliganded quadrant of the opportunity map. Both are nociceptor transcription factors. Both are also the only candidates whose tissue specificity the pipeline cannot assess — because neither GTEx nor Tabula Sapiens samples the **dorsal root ganglion**, which is where this disease lives. *PRDM12* peaks at 0.52 TPM across all 50 GTEx tissues.

The gap in the expression atlases and the gap in the literature point at the same missing tissue. That is a more useful finding than a score, and it suggests what a project on either gene would actually have to start by doing.

4. A route that emerged from the data rather than the hypothesis

Three candidates form a coherent group nobody put there on purpose: *ADCYAP1* (PACAP), its receptor *ADCYAP1R1* (PAC1), and *ADORA2A*. The literature describes GPCR transactivation of TrkA via Src-family kinases — and both *FYN* and *SRC* are in the pool, on interaction evidence. *ADCYAP1R1* is neural-enriched in **both** expression resources and already has a chemical probe.

ADORA2A knockout mice show an **increased thermal nociceptive threshold**, meaning the receptor normally promotes nociception — so activating it is the right direction. That is independent support from a different data type, which is exactly what the mouse axis was added to provide.

The same pipeline raises the objection: *ADORA2A* is **bulk only**, consistent with a predominantly striatal receptor, so whether it is present on peripheral nociceptors at all is open. Three claims in this paragraph are literature claims I have not yet checked line by line against the primary papers, and they are marked as such in the repository.

2025 versus 2026: what happened to the hand-picked genes

The exam selection was made by hand against a declared criterion: no known drugs, no chemical probe, at most three tractability assessments. Re-evaluated:

	genes	why
survive, with better reasons	*PRDM12*, *NTF3*	strong genetic evidence in a sister disease; a genuine TrkA-pathway ligand
do not survive	*AWAT2*, *FOXE3*, *LMX1B*, *MPV17*	mouse-phenotype similarity only
	SH2D2A, *HAX1*	weak or positional genetic evidence — *SH2D2A* sits within 1 Mb of *NTRK1* at 1q23
	SUCLA2	no longer associated in Open Targets 26.09

And the diagnosis of why: the 2025 criterion selects **undrugged** genes; Pharos Tdark selects **understudied** genes. All eight exam genes still associated with the disease are Tbio — none is Tdark. The criterion was not wrong. It was measuring a different axis, and the opportunity map in Step 3 exists to keep the two apart.

Where it went wrong, and how each error surfaced

I am including this section deliberately. Every one of these bugs produced output that looked entirely reasonable, and most of them would have gone into a write-up unnoticed. They are the part of the work I learned most from, and the reason the pipeline now declares its vocabularies instead of guessing them.

what went wrong	what it produced	how it surfaced	what changed
`raise_for_status()` discarded the HTTP body and retried 4xx errors	an opaque `HTTP 400` on a batched expression query, retried three times	a query-cost limit in Open Targets, invisible until the body was printed	the request helper now **shows the server's message on any 4xx and never retries it** — a malformed or too-expensive query stays malformed; expression is fetched per gene with a cache
expression page size set to 500	~17 of ~54 GTEx tissues retrieved, chosen arbitrarily by result order	single-cell rows were consuming ~467 of the 500 per gene — found by counting tissues per gene	page size 3000, cache file named after the page size, and an **explicit warning if truncation occurs**
nervous tissue identified by keyword	`cortex of kidney` counted as nervous tissue; most of the brain missed. **Every enrichment figure in that run was meaningless**	printing the actual GTEx biosample inventory	an explicitly declared list of biosample names, which **raises an error if a declared name is absent** from the response
fold-change with a zero denominator	`NaN` for the most brain-specific genes — precisely the ones the measure exists to find	*RIT2* returning no value	a 1 TPM pseudocount on numerator and denominator
wrong assumption about the cancer-hallmark vocabulary	cancer role flagged for **zero** genes, with *PIK3CA*, *BRAF* and *SRC* in the pool	the result was implausible on its face	printed the distinct attribute names and matched on them
mouse phenotype	*SORT1* scored a full	reading the matched	a declared

what went wrong	what it produced	how it surfaced	what changed
keywords too permissive	nociception match on a **retinal** phenotype	phenotype strings rather than the counts	`NOT_SOMATOSENSORY` exclusion list (15 records removed) and three graded tiers
strategy assigned by mechanistic class alone	*GDNF*, *BDNF*, *NTF4*, *NGFR*, *SORT1* all mapped to "raise signalling through TrkA" — biologically false	reading the strategy column against the biology	a declared per-gene override, written in the manifest
HGVS repeat notation unparsed	variants like `c.780GAA[1]` silently dropped from the edit-type analysis	counting parsed versus total	an added repeat-notation pattern

Two patterns run through the whole table.

Every error was found by looking at an intermediate value, not at the final output. The final rankings were plausible in every broken run. The tissue-matching bug was caught by printing an inventory; the phenotype bug by printing the matched strings; the cancer-role bug by printing distinct attribute names. The pipeline now prints those inventories as a matter of course, which is why the notebook is longer than it needs to be.

Guessing a controlled vocabulary is the single most dangerous thing in this kind of work. Four of the eight errors are the same error: assuming what a public resource calls things. The response in each case is the same — declare the terms explicitly and fail loudly when the resource no longer uses them.

One structural decision that came from a question, not a bug

An earlier version applied the manual curation by **replacing** cells that had already run. You pointed out, correctly, that this is both inconvenient (a Colab restart re-runs everything) and conceptually unclear — why overwrite an executed step instead of adding a later one?

That observation restructured the pipeline. Declared curation is now a **separate downstream step** applied on top of the automatic output, never mixed into it. Two consequences follow, and both turned out to be more valuable than the convenience: the pipeline runs linearly top to bottom with no backward edits, and the effect of every human judgement is a visible diff — 88 candidates in, 78 out, with one written reason per removal.

What the pipeline cannot know

This is the section I would want a reviewer to read first, so it is stated without hedging.

There is no experimental validation, and nothing here substitutes for it. Every proposed experiment is a proposal. Agreement between two expression atlases is robustness to the choice of resource —

nothing more. Calling it validation would be the central dishonesty available to a project of this kind, and the two-resource design exists precisely so the word is not needed.

Neither atlas samples the tissue the disease lives in. GTEx here provides central nervous tissue but no peripheral nerve and no dorsal root ganglion; Tabula Sapiens has no DRG either. The clearest symptom of the problem: *NTRK1* itself has **prostate** as its top GTEx tissue, while being 84-fold enriched in neural cell types at single-cell resolution. If the pipeline cannot place its own causal gene correctly in bulk tissue, bulk-only signals for any other candidate deserve the same suspicion.

The mouse axis is partly circular for one of the three candidate sources. Source-C genes entered because they cause sister diseases, so a pain phenotype in their knockouts is not fully independent of their selection. The axis is genuinely new information for source B — *KIDINS220*, *RAB7A*, *NTF4*, *NGFR*, *GDNF*, *NTRK2*, *ADORA2A* — which entered on interaction evidence alone. This is the limitation I am least sure how to address, and it is one of the questions in the next section.

The score measures target quality, not strategy plausibility. *SCN9A* and *SCN11A* score 10 out of 10 as targets, but the strategy they imply — restoring the excitability of surviving nociceptors — presumes that those neurons exist. In CIPA they largely never formed. This is why strategy, direction and genotype coverage are kept in columns separate from the total rather than folded into it: the total would otherwise assert something it has not measured.

Restoring the sequence does not regenerate neurons lost in development. In mice lacking both TrkA and Bax, sensory neurons survive but fail to innervate their targets. Any strategy downstream of the receptor must restore axon growth, not only survival — and no analysis here addresses developmental timing at all.

Variants are not patients. Allele frequencies are not modelled. In founder populations a handful of variants can cover a large share of disease chromosomes, which would change the genotype-coverage figures substantially. Weighting the variant spectrum by gnomAD frequency is the obvious next step and is not done.

The editing analysis is notation-level only. Edit types are read from HGVS strings. There is no PAM-site analysis, no editing window, no guide design, no off-target or efficiency assessment. The conclusion it supports is narrow and mostly negative: no single variant dominates the spectrum, so per-variant correction needs a guide per variant while gene addition is one construct for every genotype.

Declared curation is judgement, not computation. Ten candidates were removed by hand. Each removal carries one written reason, and every threshold and keyword list is recorded in the phase manifests — which makes the judgements contestable, not correct.

Three literature claims in the write-up are not yet verified line by line against the primary papers: GPCR transactivation of TrkA through Src-family kinases, the TrkA/Bax mouse result, and *ADORA2A* as a predominantly striatal receptor. They are flagged in the repository as pending.

Questions I would like your opinion on

The reason for sending this is advice, not approval. These are the five decisions I am least confident about.

1. Is the declared-curation structure the right compromise? Ten candidates were removed by hand, each with one written reason, in a step that sits downstream of everything automatic. The alternative was to encode each judgement as a rule — which would have been more elegant and, I think, less honest, since several of the reasons are mechanistic rather than algorithmic (partners of oncogenic *NTRK1* fusions, ligands of unrelated receptor tyrosine kinases linked only by STRING). Is keeping judgement visible and separate preferable to making it look computational?

2. Is the mouse-phenotype axis defensible given its partial circularity? It is the only functional, orthogonal evidence available without an experiment, and it produced the project's cleanest positive control. But for source-C genes it is partly circular, as set out above. The options I can see are: keep it as one axis for all genes and declare the circularity; score it only for source-B genes; or drop it from the score and report it as a marker, the way the Tdark label is handled. I have chosen the first and am not sure it is right.

3. Is the GPCR transactivation route worth pursuing, or is it an artefact of an interaction network? *ADCYAP1* / *ADCYAP1R1* / *ADORA2A* emerged from interaction evidence plus a mouse phenotype, with *FYN* and *SRC* in the pool as the plausible mechanism. It is the only route in the project I did not anticipate. It is also the one where I am least able to judge whether the literature supports transactivation strongly enough to build on, or whether PAC1 and A2A simply appear in many neuronal interaction datasets.

4. Should the score weight genotype coverage at all? At present it does not, which is why *NGF* scores 10 while serving roughly 4% of the variant spectrum. Reporting coverage in a separate column keeps the tension visible, but a reviewer could reasonably say that a target serving 4% of patients should not top a therapeutic ranking.

5. Is the dorsal-root-ganglion gap worth building a project around? The finding that *PRDM12* and *ZFHX2* are both the darkest candidates **and** the only unassessable ones points at a missing tissue rather than a missing target. Whether that is a genuine opportunity or simply the reason nobody works on them is the question I would most like your view on.

Where the code and data are

Repository: `rare-target-prioritization` — one consolidated Colab notebook (47 code cells, 11 explanatory text cells), a README, and `requirements.txt`. No API key is needed; every phase caches its API responses, so a restarted session resumes instead of re-querying.

```
notebooks/rare_target_prioritization.ipynb
data/raw/          dated API snapshots
data/processed/    23 tables + one manifest per phase
figures/           7 figures
requirements.txt
```

Every declared threshold, keyword list and manual decision is written to the phase manifests, so any value in the output can be traced back to the cached response that produced it and the rule that interpreted it.

To run it on another disease: change `DISEASE_QUERY` in the first cell. The disease is resolved by name search rather than a hard-coded ontology identifier, so the automatic steps transfer unchanged. The declared curation blocks do not transfer and have to be revisited — which is, I think, the correct behaviour rather than a shortcoming.

Resource versions in this run: Open Targets **26.09**; Pharos/IDG; ClinVar via NCBI E-utilities; UniProt **P04629**; AlphaFold DB **AF-P04629-F1** with AlphaMissense annotations; reference transcript **NM_002529.4**; GTEx and Tabula Sapiens baseline expression through Open Targets.

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